D-Vision-3D-Reconstructure

Memoir3D · Project Insight & Market Research

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The Core Idea

Every party generates hundreds of photos scattered across everyone's camera rolls. No single person saw the whole party. The solution: aggregate all photos from all attendees and use AI to reconstruct the event as a 3D navigable space with a timeline scrubber.

The key innovation is the **temporal layer**. Photos have EXIF timestamps. By grouping photos into 15-minute windows and training a separate 3D Gaussian Splat for each window — all anchored to the same coordinate frame from a single full-event COLMAP reconstruction — the user can scrub through time and watch the venue transform from an empty room to a packed dance floor.

Market Research: Competitor Landscape

Product	What It Does	The Gap vs. Memoir3D
Luma AI	Photo/video to Gaussian Splat. Cloud-based. Best visual quality.	Single-person capture only. No timeline, no multi-attendee workflow. Pivoted away from GS in late 2025.
Polycam	LiDAR + photo scan. 4.7★ / 540K iOS ratings. UE5 export.	Requires a deliberate scan session. No crowd-sourcing, no temporal dimension.
Scaniverse (Niantic)	On-device GS processing. Free.	Single device only. No temporal dimension. No multi-contributor workflow.
KIRI Engine	Photogrammetry-first, freemium.	No timeline, no crowd aggregation. Professional scanning tool.
RealityScan (Epic)	High-quality mesh from photos.	Professional tool. No event or social use case.

Conclusion: The timeline-scrubber over a crowd-sourced party scene is novel and commercially unoccupied. The main competitive risk is Polycam or Luma AI adding multi-user + temporal features — neither has announced this as of mid-2026.

Academic Research Landscape

The research foundation exists — the product implementation does not. This is the gap to fill.

Research / Tool	Relevance	Status

WildGaussians (NeurIPS 2024)	Handles in-the-wild uncontrolled photos with variable lighting. MIT license.	Research code only — no product
CSS: Crowd-Sourced GS (arXiv 2409.08562)	Directly addresses pose-free reconstruction from crowd-sourced imagery with inconsistent lighting.	Research only — no code released
InstantSplat (NVLabs)	Pose-free GS in seconds. Architecturally ideal for this use case.	**CUDA only + non-commercial license — cannot use**
Splatt3R	Zero-shot GS from uncalibrated image pairs.	**Non-commercial license — cannot use**
OpenSplat	Metal-native GS training. AGPLv3 license. Active community.	**USE THIS — runs on Apple Silicon**
COLMAP	Gold-standard SfM. Metal-accelerated on macOS via Homebrew.	**USE THIS — production ready**
SAM2 (Meta)	Segment Anything v2. Apache 2.0. MPS support confirmed.	**USE THIS — for person masking**

Apple Silicon Reality Check

Your Mac mini M-series (24 GB unified memory, Metal GPU) is capable but requires careful tool selection. Most research implementations assume NVIDIA CUDA and are blocked on your hardware.

Library	MPS / Metal Support	Verdict
WildGaussians	No — CUDA 11.8 explicitly required	**Blocked**
InstantSplat (NVLabs)	No — CUDA 12.1 explicitly required	**Blocked**
gsplat (official)	No — issue #163 open since April 2024, unresolved	**Blocked**
Splatfacto-W / nerfstudio GS	Broken — confirmed in issue #3290	**Blocked**
COLMAP / pycolmap	Yes — native Metal, brew install colmap, arm64 pip wheel	**Full support**
OpenSplat	Yes — Metal-native build (build from source with libtorch)	**Use this for GS**
SAM2	Yes — MPS with PYTORCH_ENABLE_MPS_FALLBACK=1	**Supported**
Depth Anything v2 Small	Yes — MPS + Core ML Neural Engine option	**Supported**
hloc (SuperPoint+SuperGlue)	Yes — MPS (slower than CUDA, retrieval pairing mitigates)	**Supported**

Performance Benchmarks — M-series Mac mini Estimates

Stage	100 photos	300 photos	Notes
COLMAP SIFT extraction	~45 sec	~2.5 min	Scales linearly, CPU-bound on Apple Silicon
COLMAP exhaustive matching	~2 min	~20 min	$O(n^2)$ bottleneck — switch to vocab tree matching above 200 photos
hloc SuperGlue matching	~8 min	~60+ min	Higher quality; use retrieval-based pairing to cap pair count
COLMAP incremental mapping	~5 min	~25 min	CPU-bound, bundle adjustment included in time
SAM2 masking (ViT-B+, MPS)	~5 min	~15 min	Approximately 3 seconds per image on MPS device
OpenSplat 30K iterations	~17 min	~17 min	Training time is roughly constant vs photo count (GPU-bound)
Total pipeline (per window)	~35 min	~80 min	Sequential; 4 time windows = 4× training time

24 GB unified memory is a meaningful advantage over dedicated VRAM. OpenSplat uses ~13–19 GB for a 300-photo indoor scene, which fits comfortably. The hard limit is ~2 million Gaussians — beyond that, OOM risk is high.

Product Design Highlights

Core User Journeys

Host: Creates event → shares QR code → guests upload → triggers reconstruction → reviews result → shares link. System warns about sparse coverage gaps before triggering GPU compute.

Viewer: Receives link → 8-second cinematic intro flythrough → free exploration → scrubs timeline to watch party evolve in 3D. No account required to view.

Contributor: Scans QR at venue or taps link in group chat → mobile browser, no app install → selects photos from camera roll → chunked upload → 'Your 27 photos are in the mix!'

3D Viewer UX Design Principles

Core principle: feel like **Google Maps 3D**, not Blender. Non-technical users already know pinch-to-zoom and drag-to-pan from maps.

- Mouse: left-drag = orbit · scroll = zoom · right-drag = pan · double-click = recenter orbit point
- Touch: one-finger drag = orbit · pinch = zoom · two-finger drag = pan
- Timeline: full-width bar, 30-min buckets, density histogram, 0.4s cross-fade between time windows
- Play button: auto-advance one window per 4 seconds — creates a time-lapse of the entire party

Progressive Loading — Never Show a Blank Screen

- 0–1s: Fetch scene.json → render timeline scrubber with loading skeletons per window
- 1–5s: Stream foundation_preview.ply (10% density, <5 MB) → user can already orbit gray point cloud
- 5–20s: Stream full foundation.ply in 2 MB HTTP range chunks → surfaces solidify progressively
- On demand: time windows loaded on scrubber interaction; ±1 window prefetched silently

Monetization Strategy

Per-Event Pricing (Consumer)

Tier	Price	Limits	Key Features
Free	$0	1 event, 50 photos	Memoir3D watermark on all exports
Starter	$9 one-time	500 photos	No watermark, 720p flythrough video export
Pro	$29 one-time	5 events, 2K photos/event	Face blur toggle, 1080p export, password protection
Wedding / Event	$79 one-time	Unlimited photos	4K export, embed code, priority processing queue

B2B Subscription — $99/month

Target: wedding photographers, event management companies, corporate comms teams. Includes: unlimited events, white-label viewer (remove branding, add their logo), client delivery portal, API access, 5 team seats. High LTV, low churn — repeat buyers making 20–50 events per year.

À La Carte Add-Ons (highest-margin upsells shown at sharing moment)

- Face blur pack: $5/event · AI highlight reel 60s: $8/event · Extended storage 5yr: $12/yr
- Rush processing (2-hour guarantee): $15/event · Custom vanity URL: $8 one-time

Realistic Timeline Summary

Scenario	MVP (M6)	Production (M7)	Notes
Optimistic	Week 26 (~6.5 mo)	Week 34 (~8.5 mo)	Everything works first try, no major blockers
Realistic ← use this	**Week 32 (~8 mo)**	**Week 40 (~10 mo)**	**Normal debugging, a few direction changes, life happens**
Pessimistic	Week 42 (~10.5 mo)	Week 52 (~13 mo)	Timeline feature requires a research pivot; COLMAP needs workarounds

The difference between optimistic and pessimistic is driven by two things: (1) how long the photo-to-Gaussian temporal association problem takes to solve, and (2) whether COLMAP performance on Apple Silicon is acceptable. Both can be validated in Phase 0 (Weeks 1–3).

The One Rule

Before writing a single line of app code, take 100 photos of your room and run the full COLMAP → OpenSplat pipeline on your Mac mini. The numbers you get — training time, quality, memory usage — will inform every product decision for the next 10 months. No planning document replaces this.

The most important test: does the .ply file render as a recognizable room in Supersplat? If yes, the foundation is proven and you can build with confidence. If no, you have discovered the core blocker before investing months of work.